

PROJECT DOCUMENTATION

Employee Attendance, Performance & HR Analytics

1. Project Overview

Employee Attendance, Performance & HR Analytics is an end-to-end data analytics portfolio project built to analyze employee-related data using MySQL Workbench and Power BI Desktop.

The project combines employee information, departments, job roles, attendance, leave records, performance evaluations, and salary history into a structured analytical solution.

The project demonstrates practical skills in:

- SQL
- MySQL database management
- Data cleaning
- Data profiling
- Data validation
- Relational data analysis
- SQL joins and aggregations
- DAX
- Power BI data modeling
- Dashboard development
- HR analytics
- Business-oriented data analysis

2. Business Problem

Organizations generate large amounts of employee data related to attendance, leave, performance, salaries, departments, job roles, and employee tenure.

Without proper analysis, it can be difficult for HR teams and managers to identify:

- Attendance patterns
- Absence trends
- Leave utilization
- Department-level workforce differences
- Employee performance patterns
- Salary differences
- Employee tenure patterns
- Potential relationships between attendance and performance
- Employees requiring further review

The purpose of this project is to transform structured employee data into an interactive Power BI dashboard that can help users explore these HR metrics.

3. Project Objectives

The main objectives of this project were to:

1. Create a relational HR analytics database using MySQL.
2. Store employee, department, job role, attendance, leave, performance, and salary information.
3. Perform data profiling to identify data-quality issues.
4. Clean and standardize the data without modifying the original raw data.
5. Perform SQL-based HR analytics.
6. Build analytical metrics for attendance, leave, performance, salary, and employee tenure.
7. Connect the cleaned MySQL data to Power BI.
8. Create a suitable Power BI data model.
9. Create DAX measures for important HR KPIs.
10. Build an interactive four-page HR analytics dashboard.
11. Present the final analysis in a format suitable for business and portfolio use.

4. Tools Used

Tool	Purpose
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MySQL Workbench	Database creation, SQL queries, data profiling, cleaning and analysis
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SQL	Data manipulation, cleaning, validation and analytical queries
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Power BI Desktop	Data modeling, DAX calculations and dashboard development
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DAX	Creation of analytical measures and KPIs
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No Python, Excel, Tableau, PostgreSQL, or SQL Server was used as part of the project.

5. Project Architecture

The project follows this workflow:

Raw HR Data



MySQL Database



Data Profiling



Data Cleaning



Clean Analytical Tables / Views



SQL Analysis



Power BI



Data Model



DAX Measures



Interactive Dashboard

6. Database Structure

The MySQL database used for the project is:

employee_hr_analytics

The database contains the following main tables:

employees

departments

job_roles

attendance

leaves

performance

salary_history

7. Database Tables

7.1 Employees

The "employees" table contains the primary employee information.

Main columns:

Column	Description
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employee_id	Unique employee identifier
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employee_name	Employee name
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gender	Employee gender
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age	Employee age
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city		Employee city
state		Employee state
department_id		Department reference
job_role_id		Job role reference
joining_date		Employee joining date
employment_status		Employee employment status
salary		Current employee salary

7.2 Departments

The "departments" table stores department information.

Column		Description
department_id		Unique department identifier
department_name		Department name

7.3 Job Roles

The "job_roles" table contains employee job-role information.

Column		Description
job_role_id		Unique job role identifier
job_role		Job role name
job_level		Employee job level

7.4 Attendance

The "attendance" table stores employee attendance records.

Column	Description
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attendance_id	Unique attendance record
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employee_id	Employee reference
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attendance_date	Attendance date
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attendance_status	Attendance status
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check_in_time	Employee check-in time
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check_out_time	Employee check-out time
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working_hours	Total working hours
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7.5 Leaves

The "leaves" table stores employee leave information.

Column	Description
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leave_id	Unique leave record
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employee_id	Employee reference
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leave_type	Type of leave
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start_date	Leave start date
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end_date	Leave end date
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leave_days | Number of leave days
leave_status | Leave approval/status

7.6 Performance

The "performance" table stores employee performance information.

Column | Description

performance_id | Unique performance record
employee_id | Employee reference
review_date | Performance review date
performance_rating | Overall performance rating
productivity_score | Productivity score
quality_score | Quality score
manager_rating | Manager rating
performance_status | Performance category/status

7.7 Salary History

The "salary_history" table stores historical salary information.

Column | Description

salary_id | Unique salary record
employee_id | Employee reference

salary | Salary amount

effective_date | Date salary became effective

salary_increment_percent | Salary increment percentage

8. Data Relationships

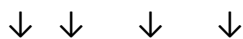
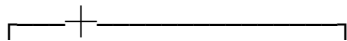
The database uses employee and reference IDs to connect the different HR datasets.

The primary relationships include:

Departments



Employees



Attendance Leaves Performance Salary History



Job Roles → Employees

The employee ID is used to connect employee records with attendance, leave, performance, and salary information.

Department and job-role IDs are used to connect employees with their respective organizational information.

9. Raw Data

The project maintains the raw data separately from the cleaned analytical data.

This was done intentionally so that:

- Original data remains available.
- Data-quality issues can be identified.
- Cleaning operations can be reviewed.
- The original dataset is not overwritten.
- The transformation process remains reproducible.

The raw data included employee-related information such as:

- Employee details
- Departments
- Job roles
- Attendance
- Leaves
- Performance
- Salary history

The dataset also contained selected data-quality issues for profiling and cleaning practice.

Examples included:

- NULL values
- Missing values
- Duplicate records
- Inconsistent capitalization
- Inconsistent text values
- Unusual working-hour values
- Unusual salary values
- Inconsistent attendance statuses
- Invalid or unusual records
- Invalid employee references

10. Data Profiling

Before cleaning the data, SQL profiling queries were used to understand the quality and structure of the raw dataset.

The profiling process checked:

- Duplicate records
- NULL values
- Missing employee information
- Inconsistent department names
- Inconsistent job-role names
- Inconsistent text formatting
- Invalid dates
- Unusual salaries

- Unusual working hours
- Unusual performance ratings
- Invalid attendance statuses
- Invalid leave records
- Invalid employee relationships

The raw tables were not overwritten during profiling.

The profiling queries are available in:

SQL/04_data_profiling.sql

11. Data Cleaning

After profiling the raw data, separate clean/analytical structures were created.

The cleaning process included:

- Handling duplicate records
- Handling NULL values
- Standardizing text values
- Standardizing department names
- Standardizing job roles
- Standardizing capitalization
- Validating dates
- Cleaning attendance records

- Cleaning leave records
- Cleaning performance records
- Cleaning salary information
- Validating employee relationships
- Calculating required analytical fields
- Preparing data for Power BI

The raw tables were kept unchanged.

The cleaning queries are available in:

SQL/05_data_cleaning.sql

12. SQL Analysis

SQL was used to perform analytical queries across the HR dataset.

The analysis covered the following areas.

Employee Analysis

- Total employees
- Active employees
- Employees by department
- Employees by job role
- Employees by city

- Employees by state
- Gender distribution
- Age distribution
- Employee tenure

Attendance Analysis

- Total attendance records
- Attendance percentage
- Absence percentage
- Work-from-home percentage
- Late attendance
- Average working hours
- Monthly attendance
- Daily attendance
- Department-wise attendance
- Employee-wise attendance
- Employees with high absence levels

Leave Analysis

- Total leave days
- Leave by type
- Leave by department
- Leave by employee
- Monthly leave trends
- Employees with high leave usage

Performance Analysis

- Average performance rating
- Department performance
- Employee performance
- Productivity score
- Quality score
- Manager rating
- High-performing employees
- Low-performing employees

Salary Analysis

- Average salary
- Salary by department
- Salary by job role
- Salary by experience
- Salary increment analysis
- Salary distribution

HR Analysis

- Employee tenure
- Employment status
- Active vs inactive employees
- Employees by job level

Combined Analysis

The project also examined relationships between:

- Attendance and performance
- Working hours and performance
- Leave and performance
- Salary and performance
- Department and performance

The complete SQL analysis is available in:

SQL/06_analytics.sql

13. Power BI Integration

After the data preparation and SQL analysis were completed, the cleaned analytical data was connected to Power BI Desktop.

The Power BI report uses the prepared MySQL data as its source.

The workflow was:

MySQL



Clean Tables / Views



Power BI



Data Model



DAX Measures



Dashboard

14. Power BI Data Model

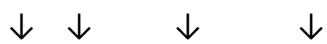
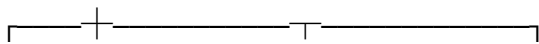
The Power BI model was designed around employee information and related HR activity.

The major entities include:

Departments



Employees



Attendance Leaves Performance Salary History

Job Roles



Employees

The model allows HR metrics to be analyzed by dimensions such as:

- Department
- Job role
- Job level
- Employee
- City
- State
- Gender
- Employment status
- Date

15. DAX Measures

DAX was used to create reusable analytical measures in Power BI.

The project includes measures for metrics such as:

- Total Employees
- Active Employees
- Total Attendance
- Present Days
- Absent Days
- Attendance Percentage
- Absence Percentage
- Total Leave Days

- Average Working Hours
- Average Performance Rating
- Average Productivity Score
- Average Quality Score
- Average Salary
- Total Salary
- Average Employee Tenure
- High Performers
- Low Performers
- Monthly Attendance
- Monthly Leave
- Monthly Performance
- Department Performance

These measures allow dashboard visuals to respond dynamically to filters and slicers.

16. Power BI Dashboard

The final Power BI report contains four analytical pages.

16.1 Page 1 — HR Executive Overview

The first page provides a high-level overview of the workforce.

The dashboard includes KPIs such as:

- Total Employees
- Active Employees
- Attendance %
- Average Performance Rating
- Average Salary
- Average Working Hours

Additional visuals include:

- Employee count by department
- Monthly attendance trend
- Employee distribution by city
- Gender distribution

This page is designed to provide a quick overview of the organization's workforce.

16.2 Page 2 — Attendance & Leave Analysis

The second page focuses on employee attendance and leave patterns.

The dashboard includes:

- Monthly attendance trend
- Attendance by department

- Attendance status
- Present vs absent
- Work-from-home records
- Late attendance
- Average working hours
- Leave by type
- Leave by department
- Monthly leave trend
- Employees with high absence levels

This page helps analyze attendance behavior and leave utilization.

16.3 Page 3 — Employee Performance

The third page focuses on employee performance.

The dashboard includes:

- Average performance rating
- Department performance
- Employee performance
- Productivity score
- Quality score
- Manager rating
- High-performing employees
- Low-performing employees

- Performance by job role
- Performance vs working hours

This page provides a way to compare employee and department performance metrics.

16.4 Page 4 — Salary & HR Analysis

The fourth page focuses on salary and broader HR metrics.

The dashboard includes:

- Average salary
- Salary by department
- Salary by job role
- Salary by experience
- Salary distribution
- Salary increment analysis
- Employee tenure
- Employment status
- High performer vs salary
- Attendance vs performance

This page combines compensation and workforce metrics for additional HR analysis.

17. Dashboard Interactivity

Power BI slicers and filters allow users to analyze the data dynamically.

Depending on the selected filters, users can analyze metrics by dimensions such as:

- Department
- Job role
- Job level
- City
- Gender
- Employment status
- Date/month

When a filter is changed, the relevant dashboard visuals update according to the selected context.

18. Key Analytical Areas

The project focuses on several important HR analytical areas.

Workforce Analysis

The dashboard can be used to understand:

- Workforce size

- Department distribution
- Job-role distribution
- Geographic distribution
- Employment status
- Employee tenure

Attendance Analysis

Attendance analysis focuses on:

- Attendance percentage
- Absence percentage
- Late attendance
- Work-from-home activity
- Working hours
- Attendance trends

Leave Analysis

Leave analysis focuses on:

- Leave volume
- Leave types
- Leave trends
- Department-level leave
- Employee leave usage

Performance Analysis

Performance analysis includes:

- Performance ratings
- Productivity
- Quality
- Manager ratings
- Department performance
- Employee performance

Compensation Analysis

Salary analysis includes:

- Average salary
- Salary distribution
- Department salary
- Job-role salary
- Salary by experience
- Salary increments

19. Key Insights

The final insights in this section should be based on the actual values shown in the completed Power BI dashboard.

Workforce

- Total employees: [Enter actual value]
- Active employees: [Enter actual value]
- Largest department: [Enter actual department]
- Most represented city: [Enter actual city]

Attendance

- Overall attendance percentage: [Enter actual value]
- Average working hours: [Enter actual value]
- Department with highest attendance: [Enter actual department]
- Department with highest absence: [Enter actual department]

Leave

- Total leave days: [Enter actual value]
- Most frequently used leave type: [Enter actual leave type]
- Department with highest leave usage: [Enter actual department]

Performance

- Average performance rating: [Enter actual value]
- Highest-performing department: [Enter actual department]
- Highest-performing job role: [Enter actual job role]

Salary

- Average salary: [Enter actual value]
- Highest-paying department: [Enter actual department]
- Highest-paying job role: [Enter actual job role]

Combined Analysis

The dashboard was also used to explore relationships between:

- Attendance and performance
- Working hours and performance
- Leave and performance
- Salary and performance

These relationships should be interpreted based on the actual dashboard results rather than assuming that one factor directly causes another.

20. Business Questions Answered

The project was designed to answer practical HR-related questions such as:

Workforce

1. How many employees are in the organization?
2. How many employees are currently active?
3. Which departments have the largest workforce?
4. Which job roles are most common?

5. How are employees distributed across cities and states?
6. What is the gender distribution?
7. What is the employee tenure distribution?

Attendance

8. What is the overall attendance percentage?
9. What is the absence percentage?
10. Which departments have higher absence levels?
11. Which employees have high absence levels?
12. What are the monthly attendance trends?
13. What is the average working time?
14. How frequently is work from home recorded?
15. How many late attendance records exist?

Leave

16. What are the most common leave types?
17. Which departments have higher leave usage?
18. Which employees use the most leave?
19. How does leave usage change over time?

Performance

20. What is the average performance rating?
21. Which departments have higher performance ratings?
22. Which job roles have higher performance?
23. How do productivity and quality scores vary?

24. How does performance compare with working hours?

Salary

25. What is the average salary?

26. How does salary vary by department?

27. How does salary vary by job role?

28. How does salary vary with employee tenure?

29. What salary increments are recorded?

30. How does salary compare with performance?

21. Data Refresh Workflow

The project follows this data-refresh workflow:

MySQL Data



Clean Tables / Views



Power BI



Refresh



Updated Dashboard

When new or updated records are added to the MySQL source, the Power BI dataset can be refreshed to retrieve the latest available data.

Examples of new data include:

- New employees
- New attendance records
- New leave records
- New performance reviews
- Updated salary information
- New monthly records

After the source data is updated, the Power BI report can be refreshed so that the visuals and DAX measures use the updated data.

22. Project File Structure

The complete project is organized as follows:

Employee-Attendance-HR-Analytics/

|

|— SQL/

| |— 01_database_setup.sql

| |— 02_create_tables.sql

| |— 03_insert_raw_data.sql

| |— 04_data_profiling.sql

- | | └─ 05_data_cleaning.sql
- | | └─ 06_analytics.sql
- |
- | └─ PowerBI/
- | | └─ Employee_Attendance_HR_Analytics.pbix
- |
- | └─ Screenshots/
- | | └─ dashboard_overview.png
- | | └─ attendance_analysis.png
- | | └─ employee_performance.png
- | | └─ salary_hr_analysis.png
- |
- | └─ Documentation/
- | | └─ Project_Documentation.md
- | | └─ Data_Dictionary.md
- | | └─ Business_Questions.md
- |
- | └─ README.md

23. How to Run the Project

Step 1 — Create the Database

Open MySQL Workbench and execute:

SQL/01_database_setup.sql

This creates the project database.

Step 2 — Create Tables

Execute:

SQL/02_create_tables.sql

This creates the relational tables and their keys.

Step 3 — Insert Raw Data

Execute:

SQL/03_insert_raw_data.sql

This populates the database with the project data.

Step 4 — Profile the Data

Execute:

SQL/04_data_profiling.sql

This identifies data-quality issues in the raw data.

Step 5 — Clean the Data

Execute:

SQL/05_data_cleaning.sql

This prepares clean analytical structures while keeping the raw data unchanged.

Step 6 — Run Analytical Queries

Execute:

SQL/06_analytics.sql

This contains the main SQL-based HR analysis.

Step 7 — Open Power BI

Open:

PowerBI/Employee_Attendance_HR_Analytics.pbix

The Power BI report contains the completed HR analytics dashboard.

24. GitHub Repository

The project is maintained in a GitHub repository containing:

- SQL scripts
- Power BI report
- Dashboard screenshots
- Project documentation
- README

The repository is structured so that another person can understand the project workflow and review the SQL and Power BI work.

25. Project Limitations

This project is a portfolio analytics project based on a prepared HR dataset.

Therefore:

- The dataset does not represent a real company's confidential employee records.
- Employee information is used for analytical and demonstration purposes.
- HR metrics should not be interpreted as actual organizational statistics.
- Correlation between HR metrics does not automatically establish causation.
- The dashboard is intended for analytical demonstration rather than production HR decision-making.

26. Future Improvements

Potential future improvements could include:

- Adding a dedicated calendar/date dimension.
- Adding more historical employee records.
- Adding recruitment analytics.
- Adding employee attrition analysis.
- Adding employee satisfaction data.
- Adding training and development metrics.
- Adding overtime analysis.
- Adding absenteeism trend forecasting.
- Adding more advanced DAX calculations.
- Adding additional dashboard pages based on business requirements.

These improvements are outside the current project scope unless implemented separately.

27. Skills Demonstrated

This project demonstrates practical experience with:

SQL

- Database creation
- Table creation
- Primary keys
- Foreign keys
- JOINS
- GROUP BY
- Aggregations
- CASE expressions
- Filtering
- Date functions
- Duplicate detection
- NULL handling
- Data cleaning
- Analytical queries

MySQL

- Database management
- Relational data modeling

- Data validation
- Data profiling
- Data transformation
- SQL script organization

Power BI

- MySQL data connection
- Data modeling
- Relationships
- Data visualization
- Slicers
- KPI cards
- Interactive dashboards
- Dashboard design

DAX

- Measures
- Aggregations
- Calculated metrics
- Percentage calculations
- Performance metrics
- Attendance metrics
- HR KPIs

28. Conclusion

The Employee Attendance, Performance & HR Analytics project demonstrates an end-to-end data analytics workflow using MySQL and Power BI.

The project starts with structured HR data, performs data profiling and cleaning in MySQL, applies SQL-based analysis, and presents the resulting metrics through an interactive Power BI dashboard.

The final solution brings together:

Database Management

+

SQL

+

Data Cleaning

+

Data Analysis

+

DAX

+

Power BI

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End-to-End HR Analytics Project

The project demonstrates how raw relational data can be transformed into meaningful analytical dashboards for exploring employee attendance, leave, performance, salary, and workforce-related metrics.